

LANSING, IL, USA

WMO: 722126

Lat: 41.540N

Lon: 87.532W

Elev: 188

StdP: 99.09

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04879

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.3	-15.4	-23.2	0.5	-17.4	-20.7	0.6	-14.9	12.3	-0.7	11.0	0.9	3.4	270	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	32.6	23.4	31.3	22.8	29.8	22.0	25.1	30.2	24.2	28.7	23.4	27.9	4.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.0	27.8	22.8	17.9	26.6	22.2	17.2	26.2	77.5	30.1	73.8	28.6	70.7	27.7	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.2	8.2	DB	-22.8	35.0	4.1	1.9	-25.8	36.4	-28.2	37.5	-30.5	38.5	-33.5	39.9	(4)
(5)			WB	-23.2	26.8	3.9	1.1	-26.0	27.6	-28.3	28.3	-30.5	28.9	-33.4	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-3.9	-2.7	3.7	9.7	15.4	20.9	23.3	22.1	18.9	11.9	5.0	-0.9	(6)	
	DBStd	10.76	6.47	6.24	6.03	5.01	5.13	3.80	3.12	2.82	4.03	5.03	5.27	5.78	(7)	
	HDD10.0	1615	431	356	213	66	9	0	0	0	0	36	166	340	(8)	
	HDD18.3	3386	688	588	454	262	123	17	2	4	41	209	401	597	(9)	
	CDD10.0	1749	1	1	19	57	176	328	413	374	267	96	15	2	(10)	
	CDD18.3	476	0	0	2	3	32	96	156	121	58	10	0	0	(11)	
	CDH23.3	4111	0	0	15	40	308	848	1415	943	463	79	0	0	(12)	
	CDH26.7	1327	0	0	2	4	93	283	518	269	141	17	0	0	(13)	
(14) Wind	WSAvg	3.4	4.2	3.9	4.0	4.1	3.3	2.9	2.5	2.3	2.6	3.4	3.9	3.9	(14)	
(15) Precipitation	PrecAvg	898	53	46	57	83	105	99	98	99	71	75	65	49	(15)	
	PrecMax	1223	129	151	125	176	191	278	247	224	239	216	170	108	(16)	
	PrecMin	521	15	13	14	34	17	24	30	22	19	22	10	8	(17)	
	PrecStd	160	31	28	29	35	43	52	47	50	48	48	41	25	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.4	16.2	25.9	27.3	32.2	33.6	35.5	32.8	32.8	28.9	21.2	15.9	(19)	
		MCWB	13.4	12.1	17.7	16.3	21.5	23.2	24.6	23.6	22.9	21.1	14.6	14.3	(20)	
	2%	DB	10.2	12.1	20.0	24.0	29.1	32.0	32.7	31.3	30.6	25.7	17.6	12.5	(21)	
		MCWB	8.5	9.7	14.5	15.8	20.4	22.7	24.2	23.0	22.4	18.7	13.2	10.7	(22)	
	5%	DB	7.4	7.9	16.7	22.1	27.0	30.0	31.3	29.5	28.0	22.7	15.5	9.4	(23)	
		MCWB	6.0	6.0	12.7	14.9	19.0	21.7	23.4	22.0	20.7	17.1	11.9	8.0	(24)	
	10%	DB	4.4	5.1	12.9	18.9	24.1	27.9	29.5	27.9	26.1	20.3	12.7	7.1	(25)	
		MCWB	3.1	3.2	9.4	12.9	18.0	20.7	22.3	21.4	19.7	16.3	9.7	5.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	13.5	18.2	18.9	23.5	25.8	26.9	25.7	24.8	22.4	16.4	14.5	(27)	
		MCDB	14.6	14.7	24.1	23.9	29.6	31.3	32.2	30.5	30.0	27.4	18.6	15.5	(28)	
	2%	WB	9.0	9.9	15.6	16.9	21.7	24.0	25.6	24.4	23.5	20.4	14.4	11.1	(29)	
		MCDB	9.7	11.9	19.0	21.9	27.0	28.9	30.8	28.6	28.2	23.6	17.3	12.1	(30)	
	5%	WB	5.9	6.4	13.1	15.5	20.3	23.0	24.5	23.6	22.3	18.6	12.1	7.9	(31)	
		MCDB	7.2	7.9	16.0	20.3	25.0	28.2	29.3	27.5	26.2	20.7	14.7	9.0	(32)	
	10%	WB	3.3	3.4	9.8	13.7	19.0	22.1	23.6	22.7	21.1	16.7	10.2	5.4	(33)	
		MCDB	4.4	4.8	13.1	19.0	23.2	26.8	28.1	26.4	24.7	19.7	12.7	6.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.3	9.0	10.7	10.9	10.7	10.3	10.1	10.7	10.0	8.6	7.2	(35)	
		MCDBR	10.0	11.7	13.7	15.3	13.7	12.3	11.9	11.9	12.7	13.3	11.8	9.6	(36)	
	5% WB	MCWBR	9.0	9.7	9.3	8.9	6.9	5.7	5.1	5.3	5.6	7.5	8.3	8.6	(37)	
		MCWBR	9.5	11.0	12.5	13.7	11.9	10.8	10.5	9.5	10.7	10.7	10.4	9.1	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.302	0.327	0.352	0.408	0.438	0.443	0.457	0.452	0.423	0.363	0.322	0.302	(40)	
		taud	2.430	2.350	2.367	2.228	2.198	2.232	2.191	2.199	2.247	2.417	2.485	2.470	(41)	
	Edn at Noon	Edn at Noon	850	876	894	862	842	836	819	811	808	826	821	820	(42)	
		Edh at Noon	78	99	109	135	143	138	143	138	122	92	74	70	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.65	2.50	3.60	4.62	5.26	6.05	6.09	5.30	4.41	2.92	1.86	1.37	(44)	
	RadStd	RadStd	0.17	0.33	0.26	0.38	0.46	0.39	0.29	0.30	0.37	0.30	0.20	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page